Year 12 Maths - Differentiation

Similar questions - Differentiation from first principles

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 $\mathrm{f}(x) = x^3 - 2x$.

(a) Use differentiation from first principles to show that $\mathrm{f}'(x) = 3x^2 - 2$. **[5]**

(b) Find the coordinates of the two points on the curve $y = \mathrm{f}(x)$ at which the gradient is 10. **[3]**

(c) Find the equation of the tangent to the curve at the point with positive x-coordinate found in part **(b)**. **[2]**

2 The points $P(1, \mathrm{g}(1))$ and $Q(1+h, \mathrm{g}(1+h))$ lie on the curve $y = \mathrm{g}(x)$, where $\mathrm{g}(x) = 2x^3 - x$.

(a) Show that the gradient of the chord PQ is $5 + 6h + 2h^2$. **[4]**

(b) Hence find $\mathrm{g}'(1)$, explaining what happens as $h \to 0$. **[2]**

(c) Calculate the gradient of the chord PQ when $h = 0.1$, and say whether it is greater than or less than $\mathrm{g}'(1)$. **[2]**

Question 1

(a) $\dfrac{\mathrm{f}(x+h)-\mathrm{f}(x)}{h} = \dfrac{(x+h)^3-2(x+h)-x^3+2x}{h} = \dfrac{3x^2h+3xh^2+h^3-2h}{h} = 3x^2+3xh+h^2-2$. As $h \to 0$, $\mathrm{f}'(x) = 3x^2-2$

(b) $3x^2-2=10$, so $x^2=4$, $x=\pm 2$. $\mathrm{f}(2)=4$, $\mathrm{f}(-2)=-4$. Points $(2,4)$ and $(-2,-4)$

(c) Gradient 10 at $(2,4)$: $y-4=10(x-2)$, so $y=10x-16$

Question 2

(a) $\mathrm{g}(1)=1$ and $\mathrm{g}(1+h)=2(1+3h+3h^2+h^3)-(1+h)=1+5h+6h^2+2h^3$. Gradient $=\dfrac{5h+6h^2+2h^3}{h} = 5+6h+2h^2$

(b) As $h \to 0$ the chord becomes the tangent at P and the gradient tends to 5, so $\mathrm{g}'(1)=5$

(c) $5+0.6+0.02=5.62$, which is greater than 5